

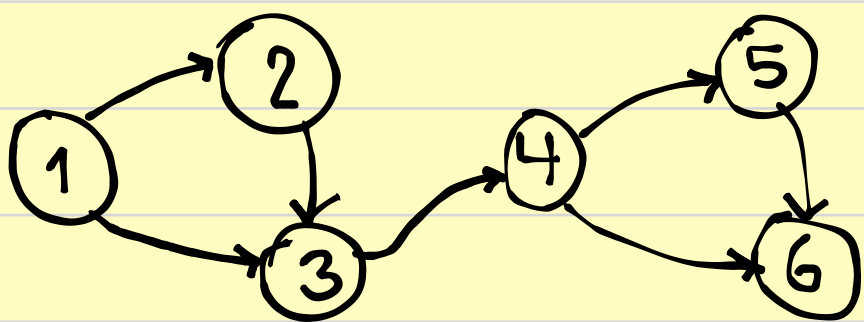
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H 4

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NETWORKS

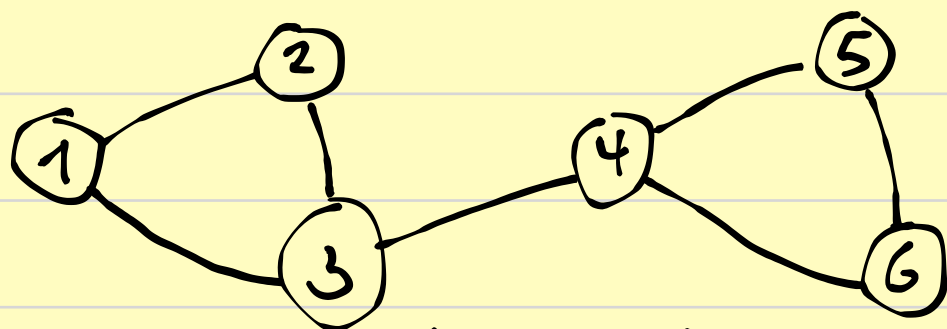
A network is defined by a group of **NODES** & **EDGES** called **GRAPH**.

$$\text{GRAPH} \equiv G(\text{Nodes \& Edges}) \equiv G(N, E)$$



$$G(N, E) = G(\{1, 2, 3, 4, 5, 6\}, \{(1 \rightarrow 2), (1 \rightarrow 3), (2 \rightarrow 3), (3 \rightarrow 4), (4 \rightarrow 5), (4 \rightarrow 6), (5 \rightarrow 6)\})$$

Direct Network when connections have a direction →

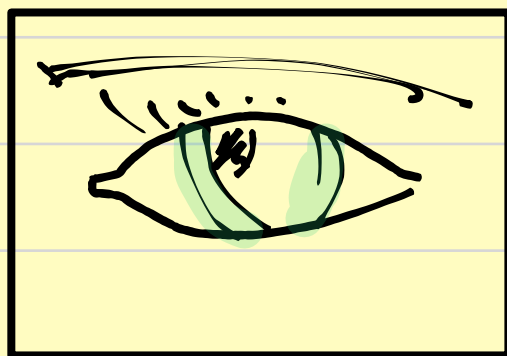


Undirected Network when connections have no direction -

A network is a non-euclidean domain.

- Euclidean domains are those in which I can measure a distance.

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$



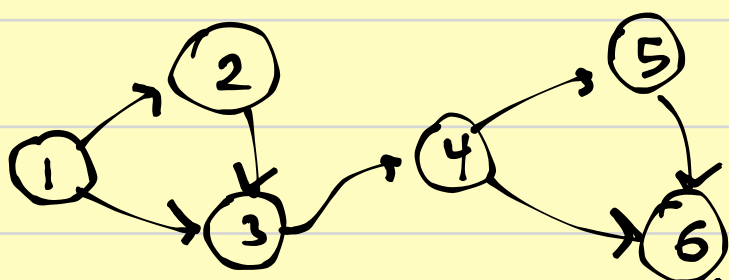
We use different KPLs to measure Network performance

- AVERAGE PATH LENGTH (APL)
- CLUSTERING COEFFICIENT (CC)
- DEGREE DISTRIBUTION (DD)
- LAPLACE MATRIX (LM)
- FIEDLER VECTOR (FV)

1. APL. Average distance btw. nodes.

$$APL = \frac{1}{N(N-1)} \sum_{i=1}^N \sum_{j=1}^N d_{ij}$$

$\nearrow$ max # connections
 $\nwarrow$ all connections happening



$$APL = \frac{1}{6(6-1)} \cdot \left(\begin{array}{l} \textcircled{1} (d_{12} \ d_{13} \ d_{14} \ d_{15} \ d_{16}) \\ \quad 1 + 1 + 2 + 3 + 3 \end{array} \right) +$$

$$\left(\begin{array}{l} \textcircled{2} (d_{21} \ d_{23} \ d_{24} \ d_{25} \ d_{26}) \\ \quad 0 + 1 + 2 + 3 + 3 \end{array} \right) +$$

$$\left(\begin{array}{l} \textcircled{3} (d_{31} \ d_{32} \ d_{34} \ d_{35} \ d_{36}) \\ \quad 0 + 0 + 1 + 2 + 2 \end{array} \right) +$$

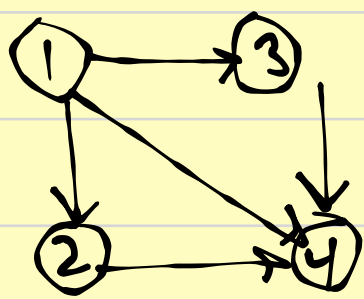
$$\left(\begin{array}{l} \textcircled{4} (d_{41} \ d_{42} \ d_{43} \ d_{45} \ d_{46}) \\ \quad 0 + 0 + 0 + 1 + 1 \end{array} \right) +$$

$$\left(\begin{array}{l} \textcircled{5} (d_{51} \ d_{52} \ d_{53} \ d_{54} \ d_{56}) \\ \quad 0 + 0 + 0 + 0 + 1 \end{array} \right) +$$

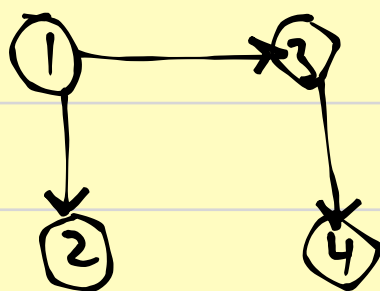
$$\left(\begin{array}{l} \textcircled{6} (d_{61} \ d_{62} \ d_{63} \ d_{64} \ d_{65}) \\ \quad 0 + 0 + 0 + 0 + 0 \end{array} \right) = \dots$$

Exercise. do the same with a non directed edges.

The best APL is the shortest.



APL₁



APL₂

If you have 2 networks with the same number of nodes, you may compare their APL. The one with the smallest will probably perform better.

2. CC. Describes how good groups are connected in the network.

$$CC = \frac{1}{N} \sum_{i=1}^N \frac{2L_i}{k_i(k_i-1)}$$

N : number of nodes
 L_i : number of relationships between the neighbours of node i

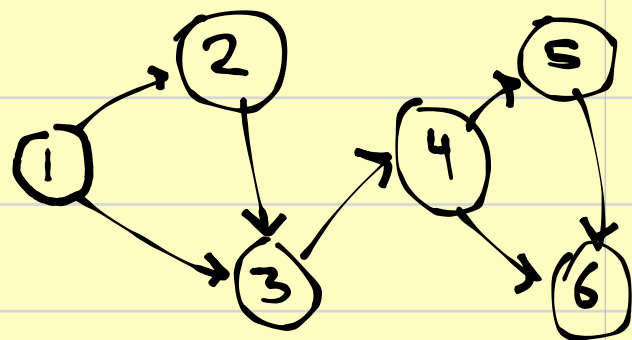
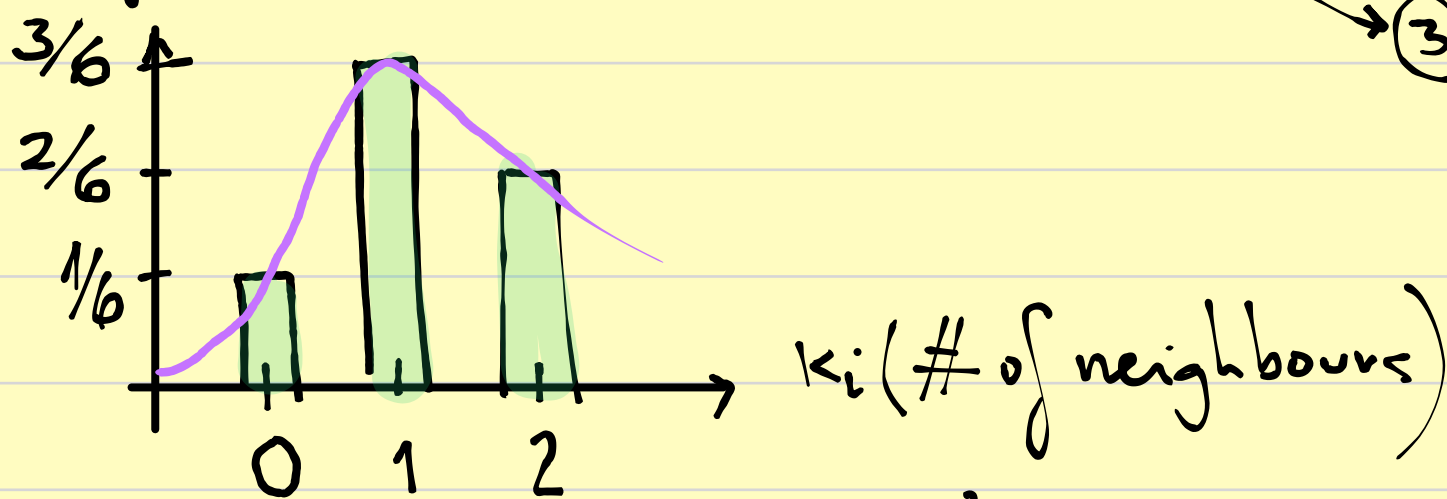
(number of my friends who are friends to each other)
 k_i = number of neighbours of node i = DEGREE of node i

$$CC = \frac{1}{6} \left[\frac{2 \cdot 1}{2 \cdot (2-1)} + \frac{2 \cdot 0}{2 \cdot 0} + \frac{2 \cdot 1}{2 \cdot (2-1)} + \frac{2 \cdot 0}{2 \cdot 0} + \frac{2 \cdot 1}{2 \cdot (2-1)} + \frac{2 \cdot 0}{2 \cdot 0} \right] = \frac{1}{3}$$

Exercise: repeat with a non-directed example above

The higher the CC, the better connected are the groups in the network.

3. Degree distribution (DD)



Ex. same network but not directed.

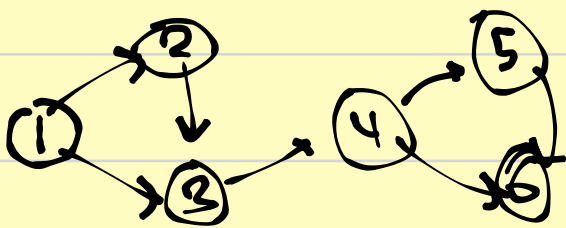
4. Laplacian Matrix (LM)

Contains relevant information of the network.

$$\mathcal{L} = D - A$$

$$D = \text{degree matrix} = \begin{cases} k_i & i=j \\ 0 & i \neq j \end{cases}$$

$$A = \text{adjacency matrix} = \begin{cases} 1 & \text{if connection} \\ 0 & \text{if no connection} \end{cases}$$



$$D = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$L = \begin{bmatrix} 2 & -1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 2 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Ex. repeat same exercise with non directed nodes.

5. FIEDLER VECTOR.

The second eigenvector of the Laplacian matrix (FV) has a sign structure that reveals bottlenecks in the network.

$$FV = \begin{bmatrix} - \\ - \\ - \\ 0 \\ + \\ + \end{bmatrix} \leftarrow \text{BOTTLENECK}$$

